Computational Linguistics/Natural Language Processing for Modern Greek: State of the Art and Prospects

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1 Introduction

Computational Linguistics and Natural Language processing are two terms that can be used interchangeably in some contexts, but can radically diverge in others. One way one can formally distinguish the two, concerns their research goals, rather than the methods used. According to this way of distinguishing between the two, Computational Linguistics tries to answer questions related to the computation of natural language by humans as these are posed by Theoretical Linguistics. It is in that sense on a par with fields like Computational Physics and/or Computational Biology. On the other hand, Natural Language Processing (NLP), especially in its Machine Learning/Deep Learning guise that has been dominating the field for close to a decade now is concerned with tasks that need to be solved.

The recent AI hype for models/systems in possession of Artificial General Intelligence (AGI), as well as the development of very fine-grained models for image recognition have brought language as a task to be solved into the forefront of AI research. Things have seen tremendous progress since then, with new neural architectures showing performance scores well-beyond what was achieved with symbolic and classic ML approaches. Models started to perform so well that they reached superhuman performance in some tasks and new more difficult benchmarks had to be constructed. An example at hand is Natural Language Understanding (NLU) tasks, traditionally considered to be very difficult to tackle. Take for instance the task of Natural Language Inference (NLI), i.e. the task of determining the logical relation between two pieces of text. Symbolic approaches to the task achieved high precision but very low recall in this task and other tasks in general (see for example Bos and Markert (2005, 2006) or can cope with the recall problem in controlled environments (Abzianidze, 2015; Mineshima et al., 2015; Bernardy and Chatzikyriakidis, 2021), but in effect they were only precise for the sentences that they had the ability to parse. However, these sentences had little variation. To the contrary, classic Machine Learning (ML) approaches had a high recall but low precision, meaning that they could

parse a variety of sentences, but were/are incapable to produce correct semantics for these sentences that would allow them to correctly decide the entailment relation (Dagan et al., 2005; Bentivogli et al., 2009). Neural Architectures managed to overcome these problems and offer human-level (sometimes superhuman) performance in this task (and various others) for most of the well-known datasets for the task. A quick look on the GLUE and SuperGLUE tables reveals this type of performance not only for NLI but for other NLU tasks (Wang et al., 2018, 2019). Here we show the performance of some models on some NLI datasets in the GLUE and SuperGLUE benchmarks as well as a table that summarizes the main characteristics of each approach:

NLI Performance on GLUE and SuperGLUE					
Model	**MNLI-m (GLUE)**	**MNLI-mm (GLUE)**	**RTE (GLUE)**	**CB (SuperGLUE)**	**RTE (SuperGLUE)**
Human Baseline	92.0	92.8	93.6	95.8/98.9	93.6
BERT-large	86.7	85.9	70.1	75.7/83.6	71.7
RoBERTa	90.8	90.2	88.2	90.5/95.2	88.2
ERNIE	92.3	91.7	92.6	98.6/99.2	92.6
T5	92.2	91.9	92.8	93.9/96.8	92.5
DeBERTa	91.9	91.6	93.2	95.0/97.0	93.2

Figure 1: Performance comparison of neural models on NLI tasks across GLUE and SuperGLUE benchmarks. Progressive improvements from BERT to more advanced architectures have enabled models to match or exceed human performance on most NLI tasks. For CB, values are shown as Accuracy/F1. Newer LLMs are not included and most probably score higher than the highest of these scores.

However, this superhuman performance of earlier transformer models has been shown to be, at least to some extent, epiphenomenal, since, increasing evidence has shown that a number of NLU datasets contain annotation artefacts and statistical irregularities that can be leveraged by machine learning models to perform well on the task (Gururangan et al., 2018; Poliak et al., 2018; Glockner et al., 2018; Talman and Chatzikyriakidis, 2019; Pham et al., 2021; Talman et al., 2021, 2022).

The advent of LLMs and especially LLM chatbots like ChatGPT, Gemini, Claude etc. has brought about a notable shift in research methodology, with increased emphasis on prompt engineering and few-shot learning approaches '(Brown et al., 2020; Zhou et al., 2022). This paradigm-shift has helped to democratize in a way access to powerful language models, but it has also raised important questions about the field's direction. LLMs' dominance has fundamentally changed the research landscape of the field, since we have moved from task-specific architectures to or towards general-purpose models are capable of adapting through careful prompt design and in-context learning (Shi et al., 2023; Sun et al., 2022; Sahoo et al., 2024; Dong et al., 2024). These models demon-

Comparison of Approaches						
Approach	**Precision**	**Recall**	**Coverage**	**Interpretability**	**Strengths**	**Limitations**
Symbolic	High	Low	Limited	High	• Precise reasoning • Explicit rules • Interpretability	• Limited variation • Brittle to changes • Manual engineering
Classical ML	Medium	High	Good	Medium	• Better generalization • Handles variations • Data-driven	• Poor semantics • Feature engineering • Limited composition
Neural	High	High	Extensive	Low	• Strong semantics • Diverse patterns • Compositional	• Low transparency • Data hungry • Dataset biases

Figure 2: Comparison of approaches for NLI tasks. Symbolic systems achieve high precision but low recall with limited coverage of language patterns. Classical ML approaches yield higher recall but medium precision. Neural architectures overcome these limitations, achieving both high precision and high recall with extensive coverage, albeit with lower interpretability.

strate remarkable capabilities across diverse linguistic tasks, yet their nature presents challenges for interpretability and systematic understanding of language processing mechanisms (Doshi-Velez and Kim, 2017; Bender et al., 2021; Lappin, 2024). Furthermore, the computational resources required for training and deploying LLMs have created new barriers to entry, potentially concentrating cutting-edge NLP research within well-funded organizations Kogkalidis and Chatzikyriakidis (2025); Lappin (2025). Lastly, serious concerns have been raised about the environmental impact, data privacy, model biases, and, crucially the sustainability of ever-larger models and a critical ongoing discussion on the devlelopment of responsible AI systems and the need for more efficient approaches to language understanding has been initiated.

This chapter is an examination of the state of Computational Linguistics and Natural Language Processing for Greek within this broader context. What I have tried to do in writing this chapter is to trace the evolution of Greek NLP through three major paradigmatic shifts: a) early symbolic approaches computationally implementing theoretical linguistic insights, b) classic machine learning approaches emphasizing feature engineering and statistical methods, and c) the current deep learning revolution that has been dominated by neural architectures and LLMs. I also discuss the crucial resources that have been developed for Greek NLP research, discuss the potential of repurposing traditional resources for NLP, and conclude by discussing general future research directions for the field. These include the potential for Neural-Symbolic (NeSy) approaches, as well as the continued relevance of traditional linguistic resources in the age of LLMs.

2 Symbolic approaches

Symbolic approaches to NLP have been very popular in the early days of NLP, following the general trend of what we nowadays sometimes call *good old fasioned AI* (GOFAI), i.e. approaches and models that dealt with computational tasks via means of symbolic systems. In Linguistics, there is a further connection with these type of models: formal models as we know them in Theoretical Linguistics (TL) are also symbolic systems, with very few exceptions. This work in TL is reflected to some extent in some of the models that have been developed in symbolic NLP, especially in tasks revolving around syntax and/or syntax/pragmatics in the sense that formal models developed in TL were computationally implemented and used for NLP tasks. It is important to note that the dominant approach to formal syntax, Generative Grammar (in most if not all of its guises), has been given less attention computationally, partly because the proponents of Generative Grammar had little interest in developing the theory to that direction, with notable exceptions, for example the excellent work by Edward Stabler (and colleagues) on the formal underpinnings of what he calls Minimalist Grammars (Stabler, 2011; Collins and Stabler, 2016; Kobele et al., 2018; Berwick and Stabler, 2019) and some other more recent attempts on using minimalist grammars on a larger scale (Torr et al., 2019; Stanojević, 2016). To my knowledge, no attempt to use Generative Grammar on a practical Greek NLP system has been made and computational approaches have been cast mainly in the family of frameworks that we call constraint-based formalisms. Some of these have counterparts in Theoretical Linguistics like Head-driven Phrase Structure Grammar (HPSG), Combinatory Categorial Grammar (CCG) and Lexical Functional Grammar (LFG), while others are just variants of mildly context-sensitive grammars.

Constraint-based formalisms is a category of formalisms used for the theoretical and/or computational study of natural languages that define a set of constraints that the languages must obey. In this sense, they are declarative in nature, rather than procedural. Typical constraint-based formalisms, as already mentioned, include HPSG (Sag et al., 1999), LFG (Dalrymple, 2001), varieties of Categorial Grammar (Moortgat, 1997; Steedman, 2001; Moot and Retoré, 2012) and various types of Unification Grammars, among other formalisms. These formalisms have been developed in contrast to the various formalisms proposed in the tradition of generative grammar. Whereas generative grammar formalisms propose a set of operations (e.g. movement, merge) that produce the set of all possible derivations (sentences) of the language, constraint-based formalisms define the constraints which determine the set of all structures that satisfy these constraints. Constraint-based formalisms have enjoyed a rigorous mathematical and computational formalization that has made them suitable to be implemented computationally.[1].

There is a lot of theoretical work on constraint-based formalisms and Greek (see the chapter ?). Here we only concentrate on constraint-based attempts

[1]For more info on the theoretical difference between the two types of grammars see (Pullum and Scholz, 2001)

that have a computational implementation. Let us start with LFG. There are a number of papers attempting the computational implementation of various syntactic phenomena in the LFG-based XLE platform. These include Multiword Expressions and MWE verbs as well as main and na-clauses (Markantonatou et al., 2019; Samaridi and Markantonatou, 2014; Fiotaki et al., 2014).

Moving on to HPSG, in Kordoni and Neu (2005), we find a grammar for Modern Greek based on HPSG and further equipped with Minimal Recursion Semantics (MRS). The grammar has been developed as part of the DELPHIN initiative and included a range of syntactic constructions and phenomena.

An interesting resource grammar developed using Ranta's Grammatical Framework (GF) is presented in Papadopoulou (2013). GF is a special-purpose programming language for developing multilingual grammar parsers and generators based on Costructive Type Theory (Martin-Löf, 1975) and functional programming. It works through parallel grammars: an abstract grammar that defines syntactic structures and concrete grammars that implement these structures through linearization rules for different languages (Ranta, 2004).

Another interesting line of work is the one presented by Michou (2007) and Michou and Seretan (2009). There, the authors use the fips multilingual parser (Wehrli, 2007), a symbolic parser that is based on a combination of ideas taken from various grammatical frameworks grammar, like GB, Minimalism, Simpler Syntax (Culicover and Jackendoff, 2005), and LFG. This is an important approach that shows the trade-off between theoretical purity and computational efficiency.

Moving away from symbolic work based on theoretical syntax, an interesting approach to developing a robust parser for Greek is presented in Boutsis et al. (2000). It uses a finite state cascade approach, applying non-recursive regular expressions to perform shallow syntactic analysis. The parser identifies phrases (e.g., noun, adjective, verb groups) and clauses using deterministic rules to ensure speed and robustness for large-scale text processing tasks. It achieves good precision and recall rates, while addressing ambiguity by enclosing unresolved structures in larger syntactic chunks. Similarly, Boutsis et al. (2000) present a symbolic Named Entity Recognition system for Greek. The system is based on a modular pipeline architecture that is built around finite-state transducers and handcrafted rules (with Brill's POS tagger being the only non-symbolic component). The system showed that a primarily symbolic approach had the potential to achieve practical performance levels by using linguistic expertise through explicit rules and lexical resources. Lastly, a good overview of a number of symbolic tools and resources developed in ILSP can be found in Prokopidis et al. (2011).

As a general note, the approaches presented here have the same strengths and weaknesses as all symbolic approaches irrespective of the specific language in case. They manage to handle difficult phenomena with precision, but can easily break down when thrown on open text or any kind of domain that requires rules they have not provisioned for. With this said, we will now move to classic Machine Learning approaches to Greek NLP. A note of caution: the reader has to keep in mind that systems in the 90s and early 00s included both symbolic

and ML-based parts, and thus the classification to purely symbolic or purely machine learning is a bit more nuanced than what we attempt here. However, this distinction can still be made on the basis of the nature of the system's core component in each case.

3 Classic Machine Learning Approaches

Classic Machine Learning (ML) has been consistently used as the dominant general approach in NLP during the 1990s and up to the mid 2010s. Contrary to what one finds in symbolic approaches, ML algorithms have the ability to learn from data, but require some sort of feature engineering to do so depending on the task. Feature engineering resembles symbolic knowledge in a way, because it attempts to encode domain knowledge that will help the model generalize over the task they need to be trained on. Feature engineering converts raw text data into informative features depending on the task. These features can be morphological (for example word stems), semantic (e.g. named entities or semantic role labelling), syntactic (dependency relations or POS tags), or very general statistical patterns (collocations, TF-IDF scores, among many others).

To give an example of how a typical classic ML system with good feature engineering works, let us look at a representative paper from that era, the one by Malakasiotis and Androutsopoulos (2007). This is a paper on Textual Entailment where the authors attempt to build a system that will classify sentences into entailments or not (this is two-way entailment, where the contradictions and the unknown categories are collapsed into one category). Here are two examples from the Recognizing Textual Entailment (RTE) dataset, one an entailment, the other a non-entailment:

(1) **T:** The drugs that slow down or halt Alzheimer's disease work best the earlier you administer them.
H: Alzheimer's disease is treated using drugs. (entailment)

(2) **T:** Drew Walker, NHS Tayside's public health director, said:"It is important to stress that this is not a confirmed case of rabies."
H: A case of rabies was confirmed. (non-entailment)

The core of the system is the linear Support Vector Machine (SVM) algorithm. SVM is a classification algorithm that uses the hyperplane to classify data into categories. The hyperplane functions as an optimal separating boundary between categories, searching for the line or plane that separates the categories in an optimal way. Simplifying a bit, the model attempts to draw a line that divides the data correctly. "Optimal" means that the line must be as far as possible from both data categories, with the goal of making the categories as distinct as possible.[2] For the case of interest, our algorithm will take as input

[2]The hyperplane is a very fundamental geometrical concept. In a geometrical space of n dimensions, the hyperplane represents an n-1 subspace of that space. This helps in generalizing the concept of a line in 2D space and the plane in 3D space.

a variety of input representations that correspond to a number of features extracted from the pair of entailment examples and will try to do the separation given these features. Here's where feature engineering comes in. What would be reasonably good features for this task? Well, let us see what the authors do. First they form 8 types of pairs based on the original pair. These are:

- Original words
- Word stems
- Part-of-speech (POS) tags
- Chunk tags
- Only nouns
- Only noun stems
- Only verbs
- Only verb stems

On these 8 pairs, ten similarity measures (e.g. Manhattan distance, Euclidean Distance and Cosine Similarity, among other measures) are used and then a score is computed that is used for the decision of the model (entailment, non-entailment). The authors use four separate SVMs, with one SVM trained specifically for each subtask of the challenge (Question Answering (QA), Information Retrieval (IR), Information Extraction (IE), Summarization (SUM)). The system achieves decent results and was one of the most prominent approaches at the RTE task back in 2007. Table 1 illustrates the specific feature sets selected for each of the four subtasks, showing how different combinations of features were tailored to optimize performance for each task.

With an understanding of how a supervised classical ML model works, we can now have a look at a number of papers using traditional ML approaches to deal with various issues in Greek NLP.

3.1 Classic ML for Greek NLP

One of the earliest approaches that uses simple ML for POS tagging is the paper by Orphanos and Christodoulakis (1999). There the authors use a decision tree algorithm paired with an elaborate lexicon to develop an elaborate POS-tag disambiguator as well as an unknown word guesser (assigns POS tags to words with no known tag). The system achieves very high accuracy. The paper is a very good example of how highly-lexicalised information can be used to facilitate training of simple ML algorithms to learn and perform in a number of fairly elaborate linguistic tasks. Papageorgiou et al. (2000) also deal with POS tagging using ML but their approach relies on using learning for the acquisition of robust symbolic rules for POS tagging. The approach contains a very elaborate tagset of POS tags (586 tags). The algorithm is a multi-tiered algorithm that

Feature sets	**#**	**IE**	**IR**	**QA**	**SUM**
similarity measures on words	10	X	X	X	X
similarity measures on stems	10	X	X	X	X
+ sim. measures on POS tags	10	X	X		
+ sim. measures on chunk tags	10	X			X
+ avg. of sim. on words (partial match)	1				X
+ avg. of sim. on stems (partial match)	1			X	X
+ avg. of sim. on POS (partial match)	1			X	X
+ avg. of sim. on chunks (partial match)	1		X		X
+ sim. on words (partial match)	10				
+ sim. on stems (partial match)	10				X
+ sim. on POS tags (partial match)	10	X			
+ sim. on chunk tags (partial match)	10				
+ negation	2	X			
+ length ratio	1	X			
+ sim. on nouns	10	X			
+ sim. on noun stems	10				
+ sim. on verbs	10				X
+ sim. on verb stems	10				
+ short/long T	1	X		X	
Total	128	64	31	23	54

Table 1: Feature selection across four subtasks. An 'X' indicates the feature was used for that subtask. The system employs different feature combinations tailored to each task, with IE using the most features (64) and QA using the fewest (23).

learns lexical as well as contextual rules to enhance POS tagging. The rules are generated iteratively and the corpus is re-tagged after each iteration. The FBT (Feature-based multi-tiered TAGGER) achieves better results than the standard TLBT (transformation-based rule tagger) achieving an error rate of 0.037 for basic POS tags (verb, noun, adjective) and increases to 0.102 when all inflectional features are taken into account. Lucarelli and Androutsopoulos (2006) present a NER system that uses a combination of SVMs and Active Learning. The system uses a pair of SVMs for the learning of person names and handcrafted rules (manually-constructed token lists) for temporal expressions. The pair of SVMs learns in two passes. The second pass uses the same machinery as the first plus the decisions made in the first pass (ability of self-correction).

While these early systems relied heavily on hand-crafted features and traditional machine learning algorithms, a significant shift in representation learning emerged in the early 2010s. The advent of word embeddings like word2vec and GloVe (Mikolov et al., 2013; Pennington et al., 2014) presented the NLP community with a wider range of capabilities with respect to semantic tasks. One paper that leverages these types of embeddings combined with traditional ML approaches is presented in Giatsoglou et al. (2017). The authors use other vector representations besides word2vec, namely lexicon-based and a combination of word2vec and lexicon representations. These representations are then used

as input for a range of classic ML algorithms, including linear kernel SVMs.[3] At this point, and before we move on to Deep Learning, it would be good to have a short digression and look at word embeddings, given that some form of these vector representations are usually used alongside DL architectures.

3.2 Word embeddings

Word embeddings are semantic representations of words in an n-dimensional space. In these representations, each word is paired with a dense vector representation capturing its meaning. The idea is that words with closer meanings will have similar vectors. A classic example that made this approach famous is the ability to perform analogy tasks. The classic example involves subtracting the vector for man from the vector for king, adding the vector for woman, and getting in return the vector for queen. Embeddings are standard input for neural models, and besides word embeddings, document embeddings also exist. Word embeddings for Greek are readily available, but can also be trained on your data, in case more specific embeddings are needed (e.g. any kind of domain-specific embeddings). Large Language Models (LLMs) learn their own embeddings that are dynamic and contextual, unlike static embeddings such as Word2Vec (Mikolov et al., 2013) or GloVe (Pennington et al., 2014). LLMs learn their embeddings through more complex architectures and are thus computationally more expensive, but also more elaborate and can be used for more complex semantic tasks, as we will see. Overall, the advent of modern embeddings has paved the way for dealing with more complex tasks in NLP compared to previous semantic representations like TF-IDF embeddings. TF-IDF embeddings represent words solely based on how they appear in documents, creating sparse vectors without capturing semantic meaning. In contrast, word embeddings generate dense vectors that learn semantic relationships between words from their contextual usage. This allows the performance of algebraic operations that reflect more fine-grained semantic relationships.

4 Deep Learning approaches

4.1 Vanilla DL architectures

The advent of neural networks in NLP during the mid-2010s has given rise to models that presented a leap in performance compared to previous approaches by a wide margin in almost all tasks. The first successful models were variations of Recurrent Neural Networks (RNNs), most specifically Long Short-Term Memory networks and Bidirectional LSTMs, which dominated the scene between 2015 and 2019. These models had a bigger memory span and are able to learn structures that needed some form of memory, but also some deeper text

[3] SVMs that perform better if the data are linearly separable, when a straight line clearly separates the different classes. Linear SVMs are also faster and, generally, overfit less than standard SVMs.

understanding. One very prominent piece of work for Greek NLP is the work presented in Prokopidis and Piperidis (2020). This presents an impressive number of neural tools for Greek NLP, including pre-trained word embeddings for Greek using FastText embeddings (Joulin et al., 2017) over a very large Greek corpus of texts, a BiLSTM-based POS tagger, a hybrid Bi-LSTM/symbolically enhanced lemmatizer, an attention model for Greek dependency parsing, and the use of FastText classifiers for the classification of Greek text. All models presented achieve very high accuracy scores. Angelidis et al. (2018) present a NER system for Greek legislative texts. The authors use various types of recurrent neural networks (including BiLSTMs and attention-based networks) to identify various types of entities, further use those for knowledge-graph enrichment, and lastly link these entities to other entities in other public datasets. Chatzikyriakidis et al. (2023) use a simple BiLSTM to classify 5 dialects of Modern Greek with accuracy over 90%.

4.2 Transformers and the LLM "revolution"

The advent of Large Language Models (LLMs) marked a new period in NLP, with stronger pre-trained models that were able to adjust to new tasks via fine-tuning and transfer learning. This development marked an important shift from previous approaches, as these models were able to use their training on vast amounts of text and their already existing understanding capabilities for other more specialised domains (transfer learning) and also efficiently acquire new understanding capabilities by fine-tuning. The first widely used LLM is BERT (Devlin et al., 2019), even though GPT-1 is the first model to attempt to leverage training on a huge corpus of unlabelled text followed by fine-tuning for various NLP tasks. GPT-1 is the first version of the GPT family of models powering the chatbot ChatGPT(Radford and Narasimhan, 2018). BERT was the first model to gain great attention in the research world, and developed into the dominant model in NLP research between 2019-2022. Naturally, BERT models for other languages have been developed throughout this period. For Modern Greek, Koutsikakis et al. (2020a) present the first monolingual Greek BERT, an important milestone for Greek NLP research, showing superiority or comparable performance to earlier non pre-trained models, as well as to Multilingual BERT models (see table 2).

Greek BERT has since been used in other tasks for Greek NLP. To give a few examples, Kotsifakou and Sotiropoulos (2023) fine-tune Greek BERT to deal with classification of political speeches, both binary and multi-class classification (recognizing between speeches of two political parties and all the parties in Greek parliament). The results show satisfactory weighted F1 scores in multi-class classification (though some parties seem to perplex the model), and very good performance in two-way classification as shown in table 3.

An important aspect of LLMs is that even in the absence of a designated LLM for a given language, not all is lost. LLMs like BERT and GPT can be used for tasks in underrepresented languages, even though not specifically trained for these languages. For example, Chatzimina et al. (2024) examine a

Model	PoS Tagging (Accuracy)	NER (Micro-F1)	NLI (Accuracy)
BILSTM-CNN-CRF	97.0 ± 0.14	76.4 ± 2.07	68.5 ± 1.71
M-BERT-UNCASED	97.8 ± 0.03	81.5 ± 1.77	73.9 ± 0.64
M-BERT-CASED	98.1 ± 0.08	82.1 ± 1.35	73.5 ± 0.49
XLM-R	**98.2 ± 0.07**	84.8 ± 1.50	77.3 ± 0.41
GREEK-BERT	**98.1 ± 0.08**	**85.7 ± 1.00**	**78.6 ± 0.62**

Table 2: Performance Results of GREEK-BERT in comparison to other models. The table shows accuracy scores for Part-of-Speech (PoS) Tagging and Natural Language Inference (NLI), and Micro-F1 for Named Entity Recognition (NER). GREEK-BERT achieves comparable or superior performance to multilingual models across all tasks, with particularly notable improvements in NER and NLI tasks. Results demonstrate the advantage of language-specific pre-training for Greek language understanding tasks.

GREEK-BERT Classification Performance Metrics			
ID	**Political Party**	**Multi-Class F1**	**Binary Classification F1-Scores**
0	Elliniki Lysi	0.89	vs. KINAL: 0.92 vs. KKE: 0.98 vs. MeRA25: 0.90 vs. ND: 0.93 vs. SYRIZA: 0.92
1	KINAL	0.67	vs. KKE: 0.95 vs. MeRA25: 0.84 vs. ND: 0.87 vs. SYRIZA: 0.76
2	KKE	0.92	vs. MeRA25: 0.94 vs. ND: 0.97 vs. SYRIZA: 0.95
3	MeRA25	0.78	vs. ND: 0.91 vs. SYRIZA: 0.75
4	Nea Dimokratia	0.82	vs. SYRIZA: 0.91
5	SYRIZA	0.56	–

Table 3: Performance metrics for GREEK-BERT political party classification. The table presents weighted F1-scores for both multi-class classification and binary classification tasks between pairs of political parties. Party names are abbreviated where appropriate: KINAL (Kinima Allagis), KKE (Kommounistiko Komma Elladas), ND (Nea Dimokratia), and SYRIZA (Synaspismos Rizospastikis Aristeras).

number of foundational models like BERT, RoBERTa, XLNet and GPT-2 for Sentiment Analysis in Greek Clinical Conversations. Surprisingly, models that are not trained specifically for Greek, BERT and RoBERTa, perform better

than multilingual models like XLNet. Mountantonakis et al. (2025) present a QA system that combines Machine Translation and BERT QA models at runtime showing very high accuracy scores. Greek BERT paved the way for other Greek specific LLMs. Greek BART (Evdaimon et al., 2024) is a sequence-to-sequence LLM that uses both an encoder and a decoder. Thus, it can be used for both discriminative and generative tasks, compared to BERT which can only do the former. The model outperforms Greek BERT in most tasks as you can see in tables 4 and 5.

Results on discriminative tasks		
Model	NCC	
	News24/7 (ours)	Makedonia (Lioudakis et al., 2020)
Greek-BERT	$92.61^{\pm 0.19}$	$89.45^{\pm 0.84}$
XLM-R	$93.1^{\pm 0.51}$	$89.6^{\pm 0.29}$
BART-random	$91.33^{\pm 0.17}$	$80.17^{\pm 0.09}$
GreekBART (ours)	$\mathbf{93.43}^{\pm 0.15}$	$\mathbf{90.12}^{\pm 0.62}$
	Sentimental Analysis	XNLI
Greek-BERT	$\mathbf{86.39}^{\pm 0.06}$	$78.6^{\pm 0.62}$
XLM-R	$85.43^{\pm 0.05}$	$78.2^{\pm 0.59}$
BART-random	$80.87^{\pm 0.12}$	$60.1^{\pm 0.43}$
GreekBART (ours)	$85.43^{\pm 0.19}$	$\mathbf{78.67}^{\pm 0.25}$

Table 4: Performance comparison of different models on discriminative tasks including News Category Classification (NCC), Sentiment Analysis, and Cross-lingual Natural Language Inference (XNLI). Results show accuracy scores with standard deviations.

Results on GreekSUM								
	GreekSUM Abstract				GreekSUM Title			
	R-1	R-2	R-L	BertScore	R-1	R-2	R-L	BertScore
LEAD	17.11	06.17	16.69	72.61/63.56	14.68	04.46	14.37	70/57.13
EXT-ORACLE	34.18	14.17	33.93	73.89/65.43	23.36	07.39	23.12	70.02/57.33
BART-random	13.85	04.47	13.65	72.44/63.27	11.55	03.27	11.42	74.47/62.22
GreekBART	**16.5**	**06.13**	**16.21**	**73.03**/64.46	15.35	05.02	15.18	75.78/63.98
mBART25	15.07	05.8	14.82	72.75/64.08	16.09	05.58	15.93	**76.81/65.38**
mBART50	15.53	06.00	15.31	**73.07/64.43**	**16.1**	**05.59**	**15.96**	**76.81/65.38**

Table 5: Evaluation of various models on the GreekSUM dataset for both abstract and title generation tasks. Metrics include ROUGE scores (R-1, R-2, R-L) and BertScore, with higher values indicating better performance.

BERT has an Ancient Greek version too. Singh et al. (2021) present an Ancient Greek BERT showing very encouraging perplexity scores and further excelling with respect to part-of-speech tags compared to other systems when tested on treebanks with Ancient or Medieval Greek data. The advent of ChatGPT and other chatbots powered by generative LLMs has sparked new research incentives and brought LLMs into the forefront of the discussion on AI. Within this context, various initiatives to train bilingual versions of LLMs for under-

represented languages have been proposed. For Greek, the first initiative of this kind is the LLM Meltemi, a bilingual model based on Mistral-7B with pretraining on 28.5B tokens of high-quality text. Llama-Krikri is the latest major Greek LLM initiative, developed by the Athena Research Center. Built on Llama-3.1-8B, it was further trained on a diverse corpus totaling 110 billion tokens, including 56.7B Greek tokens, 21B English tokens, and parallel data. Llama-Krikri achieves a +10.8 average improvement over its base model on Greek benchmarks and a +0.8 gain on English benchmarks, excelling in Greek while maintaining robust English performance (figure 6).

Performance on Greek test sets							
Model	Medical MCQA EL	Belebele EL	HellaSwag EL	ARC-Challenge EL	TruthfulQA MC2 EL	MMLU EL	Average
Meltemi 7B v1.5	42.2%	61.0%	53.8%	40.0%	49.0%	41.2%	47.9%
Llama-3.1-8B	33.4%	72.8%	52.1%	39.9%	51.1%	42.6%	48.7%
Llama-Krikri-8B (ours)	**53.8%**	**82.7%**	**64.6%**	**49.4%**	**54.2%**	**52.0%**	**59.5%**
Performance on English test sets							
Model	Winogrande	Belebele	HellaSwag	ARC-Challenge	TruthfulQA MC2	MMLU	Average
Meltemi 7B v1.5	73.4%	77.7%	79.6%	54.1%	40.5%	56.9%	63.7%
Llama-3.1-8B	**74.6%**	71.5%	**82.0%**	**58.5%**	**44.2%**	**66.2%**	**66.2%**
Llama-Krikri-8B (ours)	72.6%	**79.8%**	80.7%	55.2%	43.5%	65.0%	66.1%

Table 6: Performance comparison of different models on Greek and English language benchmarks. The Greek test sets include Medical MCQA EL, Belebele EL, HellaSwag EL, ARC-Challenge EL, TruthfulQA MC2 EL, and MMLU EL. The English test sets include Winogrande, Belebele, HellaSwag, ARC-Challenge, TruthfulQA MC2, and MMLU. Llama-Krikri-8B demonstrates an average improvement of +10.8% on Greek benchmarks while maintaining comparable performance on English benchmarks.

5 Resources for Greek NLP

One of the major issues in trying to develop computational models for natural language, especially for lower-resourced languages, is obtaining datasets that can be used to train these models. This issue has become increasingly prominent, given that the advent of neural models has produced more robust and accurate NLP systems, but for the cost of requiring vast amounts of data to be trained on. Purely symbolic approaches do not need any data to be trained, since these are models that know and do not learn. Earlier Machine Learning models need training data, but orders of magnitude less than what neural architectures require.

The question of how much data is needed is an empirical question and largely depends not only on the architecture to be trained, but also on the complexity of the task to be solved. For example, designing a language identifier between Greek and a number of its major dialects like Pontic and Cypriot is an easier task than designing a sentiment analysis or natural language inference model. Furthermore, the first task can also be dealt with using simpler Machine Learn-

ing algorithms, while sentiment analysis and NLI have been more difficult to tackle with classic ML algorithms. The first task can be solved with a dataset that contains only raw text data of the corresponding dialects, while the second requires annotation and creation of pairs of texts with their respective entailment relations. Thus, several factors affect the creation of datasets useful for training NL models:

- Ease of data collection
  - Data are available online and they can be automatically scraped, preprocessed and made ready to use.
  - Parts of the data are available online, but more need to be created (manually, automatically or semi-automatically)
  - Data have to be developed from scratch through:
    - Expert annotation
    - Crowdsourcing (online platforms like Mechanical Turk)
    - Combination of the above
- The model to be used
  - Some models would benefit from more fine-grained annotation, while others not only do not benefit but may suffer detrimental effects on performance from overly detailed annotation

To give some context on these two tasks based on hands-on experience: Chatzikyriakidis et al. (2023) shows that simple ML algorithms like Naive Bayes and Support Vector Machines (SVM) can perform reasonably well on 5-way dialect identification even with very little data. Their performance increases with more data, but already 5000 lines of each dialect are enough for performance to exceed 80%. In contrast, Kogkalidis et al. (2024) present a dataset for fine-grained entailment to test the accuracy of various LLMs that have been fine-tuned for NLI using the XNLI dataset. This dataset, containing 10k pairs of NLI examples, shows very weak results when used to evaluate these models.

There has been considerable effort in creating datasets and resources for Greek in recent years, following the general increase in NLP research. As Papantoniou and Tzitzikas (2024) documents, there were 99 NLP papers on Greek NLP and resources for the period 2015-2022 compared to 29 for the period 2000-2010. We are at a point where both Modern Greek and Ancient Greek have considerable resources of various types, while some sparse work on dialectal resources is also available. Below we mention some representative examples, since an exhaustive list is beyond the scope of this paper. For an attempt at an exhaustive list, please see Papantoniou and Tzitzikas (2024). We have also omitted resources like electronic dictionaries and glossaries. These are of course very important resources but are usually not directly usable for computational purposes. The reader is again directed to Papantoniou and Tzitzikas (2024) for these types of resources as well.

- Modern Greek: A number of datasets ranging from raw text corpora to datasets annotated for specialized tasks exist
  - Raw text: The first large corpus of Modern Greek texts is presented in Goutsos (2010). It consists of 30 million words and contains transcribed speech and written text. The Hellenic National Corpus is an impressive collection of 97 million words and has been developed and maintained at ILSP (Hatzigeorgiu et al., 2000). Both these datasets are free for research purposes, and access is granted by creating an account. A freely available dataset that also includes a lot of Greek data is the CC-100 dataset Wenzek et al. (2019). This is a massive dataset that includes around 5 billion words for Greek using Common Crawl.[4]
  - Treebanks: The Greek UD treebank is an ongoing project developed and maintained at ILSP. It consists of 2.5k UD annotated sentences.[5]
  - Specialized NLP tasks: Coreference Resolution and Question Answering datasets can be found in Institute for Language and Speech Processing (2015) and Mountantonakis et al. (2022) respectively, while textual entailment (NLI) datasets are available in Marzelou et al. (2008); Amanaki et al. (2022); Kogkalidis et al. (2024)
  - Dialectal: Here we find fewer resources. We can mention the Cypriot Greek part of the Multi-Cast corpus (Multilingual Corpus of Annotated Spoken Texts) (Haig and Schnell, 2021), the parallel POS-tagged corpus of Griko and Italian by Anastasopoulos et al. (2018), and the work by Chatzikyriakidis et al. (2023) containing raw text data mostly gathered from the web for Cypriot, Pontic, Cretan and Northern Greek.
  - Ancient Greek: Ancient Greek probably has the most resources regarding morphosyntactic annotated corpora. The Perseus Digital Library includes POS-tagged and raw text options for 13.5 million words of Ancient Greek text. Dependency treebanks can be found in Haug and Jøhndal (2008); Gorman (2020), the Diorisis corpus contains an impressive 10 million words with morphological annotation for each word (Vatri and McGillivray, 2018), and the Glaux corpus has Greek texts from the 7th century BC up to 3rd century A.D., that are syntactically annotated, POS-tagged and morphologically annotated (Keersmaekers, 2021).

The general takeaway from the situation regarding resources for Greek NLP is that there has been considerable progress in recent years in developing resources for NLP. This is the result of both individual effort and focused collaborative work by national research centres like ILSP and more general EU

[4] `https://commoncrawl.org/`

[5] `https://universaldependencies.org/treebanks/el_gdt/index.html`

initiatives like Clarin. There is still a long way to go, especially with dialectal data and more fine-grained task-oriented resources.

One last issue I want to address is the issue of data openness: there seems to be a lot of linguistic data that are/were part of research projects in Greek and other universities or institutions, that have never been made available, or have very difficult and idiosyncratic access rights. I believe it is time, given that we agree on open science principles, that these resources are made available. Nowadays, sharing our datasets and data is not only good practice, but a requirement to publish in most respected venues, so it might be a good time to try and apply these guidelines to older research retrospectively. I want to end the discussion on resources with a proposal concerning the use of traditional lexicons for NLP research, which I think has both research and practical value.

5.1 Flashback to the Future: NLP and the use of traditional language resources

The use of computational language resources has been extremely valuable throughout the history of NLP, and has been beneficial for all types of approaches, symbolic, classic ML, DL, or a combination of any of the above. However, computational resources, e.g. lexical-semantic networks or computational lexicons, are expensive and time-consuming to create. This is not difficult to understand. Given that lexicons encode years of experience from experts, they resemble traditional resources like dictionaries, in that they are fine-grained but costly to create. Traditional resources like lexicons have the expert knowledge and information needed for many NLP tasks, but their format does not allow them to be processed computationally. If the latter were possible, I do not see why one would create resources like WordNet, given that most, if not all, information in WordNet synsets appears in some way or another in a very good dictionary of English. The question is whether we are now in a position to leverage traditional resources to automatically create computational variants or subsets of these resources. Recent work suggests this is not only possible but highly effective Kogkalidis et al. (2024) propose to use a traditional dictionary, the Triantafyllides dictionary, in order to create a number of computational resources. Specifically, the authors manage to successfully use LLMs in order to extract the relevant information to create the following datasets almost fully automatically:

1. **Natural Language Inference (NLI) Dataset:**
   - 1762 sentence pairs
   - Three-way classification (Neutral, Entailment, Contradiction) + multi-label setup (more than one label possible)
   - Includes linguistic tags for diagnostic analysis
   - 1049 samples from original creation, 713 from re-annotated Greek FraCaS
2. **Word Sense Disambiguation (Greek WiC) Dataset:**

- 117,662 data points
- Binary classification task

3. **Sense Selection Dataset:**
   - 14,416 examples
   - Multi-class classification task
4. **Metaphor Detection Dataset:**
   - 3,015 examples (after filtering)
   - Binary classification task

I want to urge researchers and NLP practitioners to re-examine traditional resources and their potential to serve as high-quality datasets for NLP models. At the same time, proper digitization and processing of these resources is still very relevant even in the era of LLMs. It is a daunting task in many cases, but in the long run, and given the tools and expertise available, the reward will be great, as it will both save the NLP community valuable time from creating such resources in a computational format from scratch and allow us to train our models with higher quality data.

6 Where do we go from here?

If this chapter were written some 4 or 5 years ago, it would have been completely different. This is not only a vacuously true statement, but also true and meaningful in another way: the field is at a unique stage having reached a point of development that could not have been anticipated even by the most fervent Deep Learning enthusiasts ten years ago. At the same time, the field is in a deep crisis. A comprehensive analysis of this crisis is beyond the scope of this chapter, however, the issues I discuss below, namely the growing disconnect between computational and theoretical linguistics, the challenge to maintain linguistic diversity in the era of LLMs, and the resource constraints that lower-resourced languages are facing, describe symptoms as well as potential solutions to this crisis.

6.1 Computational Linguistics/NLP and Linguistics: Two worlds apart?

The first issue I want to draw attention to concerns the relation between Computational Linguistics/NLP on the one hand, and Theoretical Linguistics on the other. The situation can be summarized as follows: the development of stronger and more accurate NLP models has proceeded in a direction largely orthogonal to traditional theoretical linguistics (TL). While earlier NLP endeavours sought computational implementations of theoretical insights, recent research indicates a trend of gradual dissociation between the two fields. The advent of LLMs

has made this divide even greater, reaching a point where the two fields have fewer points of connection. This is a situation that we need to examine more carefully. Specifically, we need to investigate the role of Theoretical Linguistics in the era of LLMs and similarly the usefulness of LLMs for the study of Theoretical Linguistics.

The problem mostly lies in the assumptions one makes about the nature of language: Theoretical Linguistics comes equipped with theories about linguistic structure that can range from informal observations to rigorous formal systems. On the other hand, LLMs manage to produce and "understand" language in a way that disregards or totally violates basic assumptions about linguistic structure. This does not mean that theories based on these assumptions are disproven, but rather that there are alternative ways to acquire language. LLMs present one such alternative way. What is crucial in my opinion is whether (a) explicit linguistic knowledge in the form of symbolic models can enhance LLM performance, and (b) whether experimental support for different competing linguistic theories can be produced by infusing these theories as biases into LLM architectures.

The issue of reconsidering the relation between NLP and Theoretical Linguistics is of crucial importance to the study of Greek NLP. One research avenue I would like to propose is the investigation of questions relating to issues in Greek linguistics through the lens of language-specific LLMs for Greek. We are now in a position to do this, since a number of these models, as already mentioned, exist for both understanding Koutsikakis et al. (2020b); Evdaimon et al. (2024) and generation (Meltemi, Krikri).

6.2 Neural-symbolic AI

Neural-symbolic approaches are a promising avenue of research in NLP and AI in general, given that they combine the robustness of neural computation with the rigorousness and precision of symbolic computation. NeSy NLP approaches have been proposed for a variety of tasks including NLI using Neural-Symbolic Natural Logic (Feng et al., 2022), summarization with factual awareness (Wan and Bansal, 2022), textual logical reasoning (Kalyanpur et al., 2022), and question answering via a dynamic neuro-symbolic knowledge-graph dependent model (Bosselut et al., 2021).[6]

This avenue of research is worthwhile and important in the context of Greek NLP as well. The challenge is to develop hybrid NeSy systems that are able to compete in performance with purely neural models, while at the same time retaining the advantages of symbolic computation, namely explainability, transparency and, ideally, computational efficiency as well.

One promising starting avenue for such NeSy systems in Greek NLP is the use of Retrieval-Augmented Generation (RAG), i.e. systems that leverage some external source to reduce some of the well-known limitations that LLMs have,

[6]This list is far from exhaustive. The interested reader is directed to Panchendrarajan and Zubiaga (2024) and Colelough and Regli (2025) for detailed and systematic reviews of NeSy NLP.

including hallucinations and lack of systematic generalization. RAG approaches have been used with great success for various NLP tasks (Lewis et al., 2020; Izacard et al., 2023). For Greek specifically, Chatzikyriakidis and Natsina (2025) used RAG in order to check the LLMs' abilities to generate poetry in a very specific style. This external source can come in a variety of forms and there is nothing to preclude this information from being linguistic information inspired by work in Theoretical Linguistics.

Deeper integration architectures of neural and symbolic systems require more thought and care. However, using Kautz's NeSy continuum as a guide (Kautz, 2022), one can chart a practical NeSy path for Greek NLP: it begins with shallow RAG approaches that consult symbolic information, e.g. in the form of lexicon entries or morphology tables, then advances to pipelines that feed those same symbolic resources back into the model as soft prompts or auxiliary training losses, and ultimately targets fully entangled models in which a differentiable logic engine or knowledge-graph reasoner co-trains with the neural network. Each stop in this process can be used to tighten the link between computation and linguistic theory, reduce hallucinations, while at the same time allowing rival grammatical hypotheses to be encoded as explicit rules with measurable impact on downstream performance.

6.3 Working with lower-resourced languages

Greek researchers face a challenge that Kogkalidis and Chatzikyriakidis (2025) coin as "double exclusion", i.e. a general technological exclusion that divides the field into people with access to computational resources and those without access, and another exclusion that stems from the specific limitations one encounters when working with less-resourced languages. It is an objective fact that most of the benchmarks and scoreboards concern a small subset of languages. Needless to say, Greek sits outside this privileged subset of languages.

In this context, I believe that the future of Greek NLP has to concentrate on a number of important considerations. The first is resource efficiency. Given the limited computational resources that most if not all researchers of Greek NLP face, approaches that achieve good performance with fewer resources should be prioritized. This is not only a practical constraint, but also a more general prioritization of approaches that have minimal environmental impact, in line with principles of Green AI (Schwartz et al., 2020).

The next consideration is linguistic grounding. The work on multilingual LLMs that treat all languages as instances of the "just another language" approach are indeed impressive and highly practical, albeit with some problems Conneau et al. (2020); Zhang et al. (2023). However, there is a danger in these cases that might escape the usual performance metric evaluation in NLP: the loss of critical information in all layers of linguistic analysis that are specific and idiosyncratic to individual languages. I believe we might see a situation of LLM language levelling that is due to the infusion of characteristics/features of the predominant languages into the lower-resourced languages and the progressive loss of the most distinctive characteristics of lower-resourced languages. It

is, thus, imperative to strengthen and continue work on language specific NLP tasks rather than just relying on the existing LLM models and their multilingual capabilities.

7 Conclusions

In this chapter, I have presented an overview of the state of Greek NLP. I have traced its evolution from purely symbolic approaches all the way to modern approaches using Large Language Models. The general takeaway is that Greek NLP has managed to keep pace with global trends in NLP, while at the same time respecting the language-specific idiosyncrasies of Greek.

I have further presented a number of ways that I believe will move the field forward. First, revisiting traditional resources that have been computationally ignored is crucial. These resources contain expert knowledge accumulated over decades that can, at last, be leveraged with modern techniques to create high-quality computational datasets and avoid duplicating existing work when creating new computational resources. Second, we need to enhance collaboration between NLP and Theoretical Linguistics to narrow the growing gap between them. This collaboration can have bidirectional benefits: on the one hand, providing insights from linguistic theories that can enhance NLP model performance, and on the other hand, offering computational tools and methods for testing and refining linguistic theories. Retrieval-Augmented Generation and, most importantly, serious engagement with neural-symbolic systems are promising ways of combining current state-of-the-art NLP with insights from linguistic theory.

Lastly, Greek NLP must deal with the double-exclusion situation in a strategic way, by staying on top of the recent advancements in AI and multi-lingual foundational LLMs, but at the same time strive to work on language specific NLP tasks and models.

References

Abzianidze, L. (2015). A tableau prover for natural logic and language. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics (ACL).

Amanaki, E., J.-P. Bernardy, S. Chatzikyriakidis, R. Cooper, S. Dobnik, A. Karimi, A. Ek, E. C. Giannikouri, V. Katsouli, I. Kolokousis, et al. (2022). Fine-grained entailment: Resources for greek nli and precise entailment. In *Proceedings of the Workshop on Dataset Creation for Lower-Resourced Languages within the 13th Language Resources and Evaluation Conference*, pp. 44–52.

Anastasopoulos, A., M. Lekakou, J. Quer, E. Zimianiti, J. DeBenedetto, and D. Chiang (2018). Part-of-speech tagging on an endangered language: a paral-

lel griko-italian resource. In *Proceedings of the 27th International Conference on Computational Linguistics*, pp. 2529–2539.

Angelidis, I., I. Chalkidis, and M. Koubarakis (2018). Named entity recognition, linking and generation for greek legislation. In *International Conference on Legal Knowledge and Information Systems*.

Bender, E. M., T. Gebru, A. McMillan-Major, and S. Shmitchell (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*, pp. 610–623.

Bentivogli, L., P. Clark, I. Dagan, and D. Giampiccolo (2009). The fifth pascal recognizing textual entailment challenge. *TAC* 7(8), 1.

Bernardy, J.-P. and S. Chatzikyriakidis (2021). Applied temporal analysis: A complete run of the fracas test suite. In *Proceedings of the 14th International Conference on Computational Semantics (IWCS)*, pp. 11–20.

Berwick, R. C. and E. P. Stabler (2019). *Minimalist parsing*. Oxford University Press.

Bos, J. and K. Markert (2005). Recognising textual entailment with logical inference. In *Proceedings of Human Language Technology Conference and Conference on Empirical Methods in Natural Language Processing*, pp. 628–635.

Bos, J. and K. Markert (2006). When logical inference helps determining textual entailment (and when it doesn't). In *Proceedings of the second PASCAL RTE challenge*, Volume 26.

Bosselut, A., R. Le Bras, and Y. Choi (2021). Dynamic neuro-symbolic knowledge graph construction for zero-shot commonsense question answering. In *Proceedings of the AAAI conference on Artificial Intelligence*, Volume 35, pp. 4923–4931.

Boutsis, S., P. Prokopidis, V. Giouli, and S. Piperidis (2000). A robust parser for unrestricted greek text. In *LREC*. Citeseer.

Brown, T., B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al. (2020). Language models are few-shot learners. *Advances in neural information processing systems 33*, 1877–1901.

Chatzikyriakidis, S. and A. Natsina (2025, May). Poetry in RAGs: Modern Greek interwar poetry generation using RAG and contrastive training. In M. Hämäläinen, E. Öhman, Y. Bizzoni, S. Miyagawa, and K. Alnajjar (Eds.), *Proceedings of the 5th International Conference on Natural Language Processing for Digital Humanities*, Albuquerque, USA, pp. 257–264. Association for Computational Linguistics.

Chatzikyriakidis, S., C. Qwaider, I. Kolokousis, C. Koula, D. Papadakis, and E. Sakellariou (2023). Grdd: A dataset for greek dialectal nlp.

Chatzimina, M. E., H. A. Papadaki, C. Pontikoglou, and M. Tsiknakis (2024). A comparative sentiment analysis of greek clinical conversations using bert, roberta, gpt-2, and xlnet. *Bioengineering 11*(6), 521.

Colelough, B. C. and W. Regli (2025). Neuro-symbolic ai in 2024: A systematic review. *arXiv preprint arXiv:2501.05435*.

Collins, C. and E. Stabler (2016). A formalization of minimalist syntax. *Syntax 19*(1), 43–78.

Conneau, A., K. Khandelwal, N. Goyal, V. Chaudhary, G. Wenzek, F. Guzmán, E. Grave, M. Ott, L. Zettlemoyer, and V. Stoyanov (2020, July). Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Online, pp. 8440–8451. Association for Computational Linguistics.

Culicover, P. W. and R. Jackendoff (2005). *Simpler Syntax*. Oxford University Press UK.

Dagan, I., O. Glickman, and B. Magnini (2005). The pascal recognising textual entailment challenge. In *Machine learning challenges workshop*, pp. 177–190. Springer.

Dalrymple, M. (2001). *Lexical functional grammar*, Volume 34. Brill.

Devlin, J., M.-W. Chang, K. Lee, and K. Toutanova (2019). Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pp. 4171–4186.

Dong, Q., L. Li, D. Dai, C. Zheng, J. Ma, R. Li, H. Xia, J. Xu, Z. Wu, B. Chang, et al. (2024). A survey on in-context learning. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 1107–1128.

Doshi-Velez, F. and B. Kim (2017). Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608*.

Evdaimon, I., H. Abdine, C. Xypolopoulos, S. Outsios, M. Vazirgiannis, and G. Stamou (2024). Greekbart: The first pretrained greek sequence-to-sequence model. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pp. 7949–7962.

Feng, Y., X. Yang, X. Zhu, and M. Greenspan (2022). Neuro-symbolic natural logic with introspective revision for natural language inference. *Transactions of the Association for Computational Linguistics 10*, 240–256.

Fiotaki, A., S. Markantonatou, M. Butt, and T. H. King (2014). Modern greek tense in main and na subordinated clauses: An lfg/xle treatment. In *Proceedings of LFG14*.

Giatsoglou, M., M. G. Vozalis, K. Diamantaras, A. Vakali, G. Sarigiannidis, and K. C. Chatzisavvas (2017). Sentiment analysis leveraging emotions and word embeddings. *Expert Systems with Applications 69*, 214–224.

Glockner, M., V. Shwartz, and Y. Goldberg (2018). Breaking nli systems with sentences that require simple lexical inferences. In *Proceedings of ACL*.

Gorman, V. B. (2020). Dependency treebanks of ancient greek prose. *Journal of Open Humanities Data 6*(1).

Goutsos, D. (2010). The corpus of greek texts: a reference corpus for modern greek. *Corpora 5*(1), 29–44.

Gururangan, S., S. Swayamdipta, O. Levy, R. Schwartz, S. Bowman, and N. A. Smith (2018, June). Annotation artifacts in natural language inference data. In *Proceedings of NAACL*, New Orleans, Louisiana, pp. 107–112. Association for Computational Linguistics.

Haig, G. and S. Schnell (2021). Multi-cast: Multilingual corpus of annotated spoken texts, version 2108.

Hatzigeorgiu, N., M. Gavrilidou, S. Piperidis, G. Carayannis, A. Papakostopoulou, A. Spiliotopoulou, A. Vacalopoulou, P. Labropoulou, E. Mantzari, H. Papageorgiou, et al. (2000). Design and implementation of the online ilsp greek corpus. In *LREC*.

Haug, D. T. and M. Jøhndal (2008). Creating a parallel treebank of the old indo-european bible translations. In *Proceedings of the second workshop on language technology for cultural heritage data (LaTeCH 2008)*, pp. 27–34.

Institute for Language and Speech Processing (2015). Institute for Language and Speech Processing Corpus 752. `https://inventory.clarin.gr/corpus/752`. [Online; accessed 15-November-2024].

Izacard, G., P. Lewis, M. Lomeli, L. Hosseini, F. Petroni, T. Schick, J. Dwivedi-Yu, A. Joulin, S. Riedel, and E. Grave (2023). Atlas: Few-shot learning with retrieval augmented language models. *Journal of Machine Learning Research 24*(251), 1–43.

Joulin, A., E. Grave, and P. B. T. Mikolov (2017). Bag of tricks for efficient text classification. *EACL 2017*, 427.

Kalyanpur, A., T. Breloff, and D. A. Ferrucci (2022). Braid: Weaving symbolic and neural knowledge into coherent logical explanations. In *Proceedings of the AAAI conference on artificial intelligence*, Volume 36, pp. 10867–10874.

Kautz, H. (2022). The third ai summer: Aaai robert s. engelmore memorial lecture. *Ai magazine 43*(1), 105–125.

Keersmaekers, A. (2021). The glaux corpus: methodological issues in designing a long-term, diverse, multi-layered corpus of ancient greek. In *Proceedings of the 2nd International Workshop on Computational Approaches to Historical Language Change 2021*, pp. 39–50.

Kobele, G. et al. (2018). Constraining mgbank: Agreement, l-selection and supertagging in minimalist grammars. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Kogkalidis, K. and S. Chatzikyriakidis (2025, May). On tables with numbers, with numbers. In D. Nguyen (Ed.), *Proceedings of the 1st Workshop on Language Models for Underserved Communities (LM4UC 2025)*, Albuquerque, New Mexico, pp. 104–115. Association for Computational Linguistics.

Kogkalidis, K., S. Chatzikyriakidis, E. Giannikouri, V. Katsouli, C. Klironomou, C. Koula, D. Papadakis, T. Pasparaki, E. Psaltaki, E. Sakellariou, and C. Soupiona (2024, March). OYXOY: A Modern NLP test suite for Modern Greek. In Y. Graham and M. Purver (Eds.), *Findings of the Association for Computational Linguistics: EACL 2024*, St. Julian's, Malta, pp. 311–322. Association for Computational Linguistics.

Kordoni, V. and J. Neu (2005). Deep analysis of modern greek. In K.-Y. Su, J. Tsujii, J.-H. Lee, and O. Y. Kwong (Eds.), *Natural Language Processing – IJCNLP 2004*, Berlin, Heidelberg, pp. 674–683. Springer Berlin Heidelberg.

Kotsifakou, K. M. and D. N. Sotiropoulos (2023). Greek political speech classification using bert. In *2023 14th International Conference on Information, Intelligence, Systems & Applications (IISA)*, pp. 1–7. IEEE.

Koutsikakis, J., I. Chalkidis, P. Malakasiotis, and I. Androutsopoulos (2020a). Greek-bert: The greeks visiting sesame street. In *11th Hellenic conference on artificial intelligence*, pp. 110–117.

Koutsikakis, J., I. Chalkidis, P. Malakasiotis, and I. Androutsopoulos (2020b). Greek-BERT: The Greeks visiting sesame street. In *11th Hellenic Conference on Artificial Intelligence*, SETN 2020, New York, NY, USA, pp. 110–117. Association for Computing Machinery.

Lappin, S. (2024). Assessing the strengths and weaknesses of large language models. *Journal of Logic, Language and Information 33*(1), 9–20.

Lappin, S. (2025). *Understanding the Artificial Intelligence Revolution: Between Catastrophe and Utopia*. CRC Press.

Lewis, P., E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, et al. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems 33*, 9459–9474.

Lucarelli, G. and I. Androutsopoulos (2006). A greek named-entity recognizer that uses support vector machines and active learning. In *Hellenic Conference on Artificial Intelligence*, pp. 203–213. Springer.

Malakasiotis, P. and I. Androutsopoulos (2007, June). Learning textual entailment using SVMs and string similarity measures. In S. Sekine, K. Inui, I. Dagan, B. Dolan, D. Giampiccolo, and B. Magnini (Eds.), *Proceedings of the ACL-PASCAL Workshop on Textual Entailment and Paraphrasing*, Prague, pp. 42–47. Association for Computational Linguistics.

Markantonatou, S., N. Samaridi, and P. Minos (2019). Issues in parsing mwes in an lfg/xle framework. In Y. Parmentier and J. Waszczuk (Eds.), *Representation and parsing of multiword expressions*. Language Science Press.

Martin-Löf, P. (1975). An intuitionistic theory of types: Predicative part. In *Studies in Logic and the Foundations of Mathematics*, Volume 80, pp. 73–118. Elsevier.

Marzelou, E., M. Zourari, V. Giouli, and S. Piperidis (2008). Building a greek corpus for textual entailment. In *LREC*.

Michou, A. (2007, June). Analyse syntaxique et traitement automatique du syntagme nominal grec moderne. In N. Hathout and P. Muller (Eds.), *Actes de la 14ème conférence sur le Traitement Automatique des Langues Naturelles. Posters*, Toulouse, France, pp. 203–212. ATALA.

Michou, A. and V. Seretan (2009). A tool for multi-word expression extraction in modern greek using syntactic parsing. In *Proceedings of the Demonstrations Session at EACL 2009*, pp. 45–48.

Mikolov, T., I. Sutskever, K. Chen, G. S. Corrado, and J. Dean (2013). Distributed representations of words and phrases and their compositionality. *Advances in neural information processing systems 26*.

Mineshima, K., P. Martínez-Gómez, Y. Miyao, and D. Bekki (2015). Higher-order logical inference with compositional semantics. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pp. 2055–2061.

Moortgat, M. (1997). Categorial type logics. In *Handbook of logic and language*, pp. 93–177. Elsevier.

Moot, R. and C. Retoré (2012). *The logic of categorial grammars: a deductive account of natural language syntax and semantics*, Volume 6850. Springer.

Mountantonakis, M., M. Bastakis, L. Mertzanis, and Y. Tzitzikas (2022). Tiresias: Bilingual question answering over dbpedia. In *Workshop at ISWC*.

Mountantonakis, M., L. Mertzanis, M. Bastakis, and Y. Tzitzikas (2025). A comparative evaluation for question answering over greek texts by using machine translation and bert. *Language Resources and Evaluation 59*(2), 931–957.

Orphanos, G. S. and D. Christodoulakis (1999). Pos disambiguation and unknown word guessing with decision trees. In *Ninth Conference of the European Chapter of the Association for Computational Linguistics*, pp. 134–141.

Panchendrarajan, R. and A. Zubiaga (2024). Synergizing machine learning & symbolic methods: A survey on hybrid approaches to natural language processing. *Expert Systems with Applications 251*, 124097.

Papadopoulou, I. (2013). Gf modern greek resource grammar. In *Proceedings of the Student Research Workshop associated with RANLP 2013*, pp. 126–133.

Papageorgiou, H., P. Prokopidis, V. Giouli, and S. Piperidis (2000). A unified pos tagging architecture and its application to greek. In *LREC*. Citeseer.

Papantoniou, K. and Y. Tzitzikas (2024). Nlp for the greek language: A longer survey.

Pennington, J., R. Socher, and C. D. Manning (2014). Glove: Global vectors for word representation. In *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, pp. 1532–1543.

Pham, T., T. Bui, L. Mai, and A. Nguyen (2021). Out of order: How important is the sequential order of words in a sentence in natural language understanding tasks? In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pp. 1145–1160.

Poliak, A., J. Naradowsky, A. Haldar, R. Rudinger, and B. Van Durme (2018, June). Hypothesis only baselines in natural language inference. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics*, New Orleans, Louisiana, pp. 180–191. Association for Computational Linguistics.

Prokopidis, P., B. Georgantopoulos, and H. Papageorgiou (2011). A suite of natural language processing tools for greek. In *The 10th international conference of Greek linguistics*.

Prokopidis, P. and S. Piperidis (2020). A neural nlp toolkit for greek. In *11th Hellenic Conference on Artificial Intelligence*, pp. 125–128.

Pullum, G. K. and B. C. Scholz (2001). On the distinction between model-theoretic and generative-enumerative syntactic frameworks. In *International Conference on Logical Aspects of Computational Linguistics*, pp. 17–43. Springer.

Radford, A. and K. Narasimhan (2018). Improving language understanding by generative pre-training.

Ranta, A. (2004). Grammatical framework. *Journal of Functional Programming 14*(2), 145–189.

Sag, I. A., T. Wasow, E. M. Bender, and I. A. Sag (1999). *Syntactic theory: A formal introduction*, Volume 92. Center for the Study of Language and Information Stanford, CA.

Sahoo, P., A. K. Singh, S. Saha, V. Jain, S. Mondal, and A. Chadha (2024). A systematic survey of prompt engineering in large language models: Techniques and applications. *arXiv preprint arXiv:2402.07927*.

Samaridi, N. and S. Markantonatou (2014). Parsing modern greek verb mwes with lfg/xle grammars. In *Proceedings of the 10th Workshop on Multiword Expressions (MWE)*, pp. 33–37.

Schwartz, R., J. Dodge, N. A. Smith, and O. Etzioni (2020). Green ai. *Communications of the ACM 63*(12), 54–63.

Shi, C., Y. Su, C. Yang, Y. Yang, and D. Cai (2023). Specialist or generalist? instruction tuning for specific nlp tasks. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 15336–15348.

Singh, P., G. Rutten, and E. Lefever (2021). A pilot study for bert language modelling and morphological analysis for ancient and medieval greek. In *Proceedings of the 5th Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature*, pp. 128–137.

Stabler, E. P. (2011). Computational perspectives on minimalism. *Oxford Handbooks Online*.

Stanojević, M. (2016). Minimalist grammar transition-based parsing. In *International Conference on Parsing Technologies*, pp. 17–29.

Steedman, M. (2001). *The syntactic process*. MIT press.

Sun, T.-X., X.-Y. Liu, X.-P. Qiu, and X.-J. Huang (2022). Paradigm shift in natural language processing. *Machine Intelligence Research 19*(3), 169–183.

Talman, A., M. Apidianaki, S. Chatzikyriakidis, and J. Tiedemann (2021, May 31–2 June). NLI Data Sanity Check: Assessing the Effect of Data Corruption on Model Performance. In *Proceedings of the 23rd Nordic Conference on Computational Linguistics (NoDaLiDa)*, Reykjavik, Iceland (Online), pp. 276–287. Linköping University Electronic Press, Sweden.

Talman, A., M. Apidianaki, S. Chatzikyriakidis, and J. Tiedemann (2022). How does data corruption affect natural language understanding models? a study on glue datasets. In *Proceedings of the 11th Joint Conference on Lexical and Computational Semantics*, pp. 226–233.

Talman, A. and S. Chatzikyriakidis (2019, August). Testing the generalization power of neural network models across NLI benchmarks. In *Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, Florence, Italy, pp. 85–94. Association for Computational Linguistics.

Torr, J., M. Stanojević, M. Steedman, and S. B. Cohen (2019). Wide-coverage neural a* parsing for minimalist grammars. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 2357–2367.

Vatri, A. and B. McGillivray (2018). The diorisis ancient greek corpus: Linguistics and literature. *Research Data Journal for the Humanities and Social Sciences 3*(1), 55–65.

Wan, D. and M. Bansal (2022). Factpegasus: Factuality-aware pre-training and fine-tuning for abstractive summarization. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 1010–1028.

Wang, A., Y. Pruksachatkun, N. Nangia, A. Singh, J. Michael, F. Hill, O. Levy, and S. Bowman (2019). SuperGLUE: A stickier benchmark for general-purpose language understanding systems. *Advances in neural information processing systems 32*.

Wang, A., A. Singh, J. Michael, F. Hill, O. Levy, and S. Bowman (2018). Glue: A multi-task benchmark and analysis platform for natural language understanding. In *Proceedings of the 2018 EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pp. 353–355.

Wehrli, E. (2007, June). Fips, a "deep" linguistic multilingual parser. In T. Baldwin, M. Dras, J. Hockenmaier, T. H. King, and G. van Noord (Eds.), *ACL 2007 Workshop on Deep Linguistic Processing*, Prague, Czech Republic, pp. 120–127. Association for Computational Linguistics.

Wenzek, G., M.-A. Lachaux, A. Conneau, V. Chaudhary, F. Guzmán, A. Joulin, and E. Grave (2019). Ccnet: Extracting high quality monolingual datasets from web crawl data. *arXiv preprint arXiv:1911.00359*.

Zhang, X., S. Li, B. Hauer, N. Shi, and G. Kondrak (2023). Don't trust chatgpt when your question is not in english: A study of multilingual abilities and types of llms. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 7915–7927.

Zhou, Y., A. I. Muresanu, Z. Han, K. Paster, S. Pitis, H. Chan, and J. Ba (2022). Large language models are human-level prompt engineers. In *The Eleventh International Conference on Learning Representations*.